Assignment_4_Pharmaceutical_Sales

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```
# Import the pharmaceutical dataset
pharma <- read.csv("~/Downloads/Pharmaceuticals.csv")

# View the data
head(pharma)
```

```
##   Symbol                Name Market_Cap Beta PE_Ratio  ROE  ROA Asset_Turnover
## 1    ABT Abbott Laboratories      68.44 0.32     24.7 26.4 11.8            0.7
## 2    AGN      Allergan, Inc.       7.58 0.41     82.5 12.9  5.5            0.9
## 3    AHM        Amersham plc       6.30 0.46     20.7 14.9  7.8            0.9
## 4    AZN     AstraZeneca PLC      67.63 0.52     21.5 27.4 15.4            0.9
## 5    AVE             Aventis      47.16 0.32     20.1 21.8  7.5            0.6
## 6    BAY           Bayer AG       16.90 1.11     27.9  3.9  1.4            0.6
##   Leverage Rev_Growth Net_Profit_Margin Median_Recommendation Location Exchange
## 1     0.42       7.54              16.1          Moderate Buy       US     NYSE
## 2     0.60       9.16               5.5          Moderate Buy   CANADA     NYSE
## 3     0.27       7.05              11.2            Strong Buy       UK     NYSE
## 4     0.00      15.00              18.0         Moderate Sell       UK     NYSE
## 5     0.34      26.81              12.9          Moderate Buy   FRANCE     NYSE
## 6     0.00      -3.17               2.6                  Hold  GERMANY     NYSE
```

```
# Examine dataset structure
str(pharma)
```

```
## 'data.frame':    21 obs. of  14 variables:
##  $ Symbol               : chr  "ABT" "AGN" "AHM" "AZN" ...
##  $ Name                 : chr  "Abbott Laboratories" "Allergan, Inc." "Amersham plc" "AstraZeneca PL
##  $ Market_Cap           : num  68.44 7.58 6.3 67.63 47.16 ...
##  $ Beta                 : num  0.32 0.41 0.46 0.52 0.32 1.11 0.5 0.85 1.08 0.18 ...
##  $ PE_Ratio             : num  24.7 82.5 20.7 21.5 20.1 27.9 13.9 26 3.6 27.9 ...
##  $ ROE                  : num  26.4 12.9 14.9 27.4 21.8 3.9 34.8 24.1 15.1 31 ...
##  $ ROA                  : num  11.8 5.5 7.8 15.4 7.5 1.4 15.1 4.3 5.1 13.5 ...
##  $ Asset_Turnover       : num  0.7 0.9 0.9 0.9 0.6 0.6 0.9 0.6 0.3 0.6 ...
##  $ Leverage             : num  0.42 0.6 0.27 0 0.34 0 0.57 3.51 1.07 0.53 ...
##  $ Rev_Growth           : num  7.54 9.16 7.05 15 26.81 ...
##  $ Net_Profit_Margin    : num  16.1 5.5 11.2 18 12.9 2.6 20.6 7.5 13.3 23.4 ...
##  $ Median_Recommendation: chr  "Moderate Buy" "Moderate Buy" "Strong Buy" "Moderate Sell" ...
##  $ Location             : chr  "US" "CANADA" "UK" "UK" ...
##  $ Exchange             : chr  "NYSE" "NYSE" "NYSE" "NYSE" ...
```

```r
# Display summary statistics
summary(pharma)
```

```
##     Symbol              Name             Market_Cap          Beta
##  Length:21          Length:21          Min.   :  0.41   Min.   :0.1800
##  Class :character   Class :character   1st Qu.:  6.30   1st Qu.:0.3500
##  Mode  :character   Mode  :character   Median : 48.19   Median :0.4600
##                                        Mean   : 57.65   Mean   :0.5257
##                                        3rd Qu.: 73.84   3rd Qu.:0.6500
##                                        Max.   :199.47   Max.   :1.1100
##     PE_Ratio          ROE             ROA        Asset_Turnover    Leverage
##  Min.   : 3.60   Min.   : 3.9   Min.   : 1.40   Min.   :0.3    Min.   :0.0000
##  1st Qu.:18.90   1st Qu.:14.9   1st Qu.: 5.70   1st Qu.:0.6    1st Qu.:0.1600
##  Median :21.50   Median :22.6   Median :11.20   Median :0.6    Median :0.3400
##  Mean   :25.46   Mean   :25.8   Mean   :10.51   Mean   :0.7    Mean   :0.5857
##  3rd Qu.:27.90   3rd Qu.:31.0   3rd Qu.:15.00   3rd Qu.:0.9    3rd Qu.:0.6000
##  Max.   :82.50   Max.   :62.9   Max.   :20.30   Max.   :1.1    Max.   :3.5100
##    Rev_Growth    Net_Profit_Margin Median_Recommendation   Location
##  Min.   :-3.17   Min.   : 2.6      Length:21             Length:21
##  1st Qu.: 6.38   1st Qu.:11.2      Class :character      Class :character
##  Median : 9.37   Median :16.1      Mode  :character      Mode  :character
##  Mean   :13.37   Mean   :15.7
##  3rd Qu.:21.87   3rd Qu.:21.1
##  Max.   :34.21   Max.   :25.5
##    Exchange
##  Length:21
##  Class :character
##  Mode  :character
##
##
##
```

```r
# Select the 9 numerical financial variables
pharma_numeric <- pharma[, 3:11]

# View the selected data
head(pharma_numeric)
```

```
##   Market_Cap Beta PE_Ratio  ROE  ROA Asset_Turnover Leverage Rev_Growth
## 1      68.44 0.32     24.7 26.4 11.8            0.7     0.42       7.54
## 2       7.58 0.41     82.5 12.9  5.5            0.9     0.60       9.16
## 3       6.30 0.46     20.7 14.9  7.8            0.9     0.27       7.05
## 4      67.63 0.52     21.5 27.4 15.4            0.9     0.00      15.00
## 5      47.16 0.32     20.1 21.8  7.5            0.6     0.34      26.81
## 6      16.90 1.11     27.9  3.9  1.4            0.6     0.00      -3.17
##   Net_Profit_Margin
## 1              16.1
## 2               5.5
## 3              11.2
## 4              18.0
## 5              12.9
## 6               2.6
```

```r
# Standardize the financial variables
pharma_scaled <- scale(pharma_numeric)

# View standardized values
head(pharma_scaled)
```

```
##      Market_Cap        Beta    PE_Ratio         ROE        ROA Asset_Turnover
## [1,]  0.1840960 -0.80125356 -0.04671323  0.04009035  0.2416121  -5.121077e-16
## [2,] -0.8544181 -0.45070513  3.49706911 -0.85483986 -0.9422871   9.225312e-01
## [3,] -0.8762600 -0.25595600 -0.29195768 -0.72225761 -0.5100700   9.225312e-01
## [4,]  0.1702742 -0.02225704 -0.24290879  0.10638147  0.9181259   9.225312e-01
## [5,] -0.1790256 -0.80125356 -0.32874435 -0.26484883 -0.5664461  -4.612656e-01
## [6,] -0.6953818  2.27578267  0.14948233 -1.45146000 -1.7127612  -4.612656e-01
##        Leverage Rev_Growth Net_Profit_Margin
## [1,] -0.2120979 -0.5277675        0.06168225
## [2,]  0.0182843 -0.3811391       -1.55366706
## [3,] -0.4040831 -0.5721181       -0.68503583
## [4,] -0.7496565  0.1474473        0.35122600
## [5,] -0.3144900  1.2163867       -0.42597037
## [6,] -0.7496565 -1.4971443       -1.99560225
```

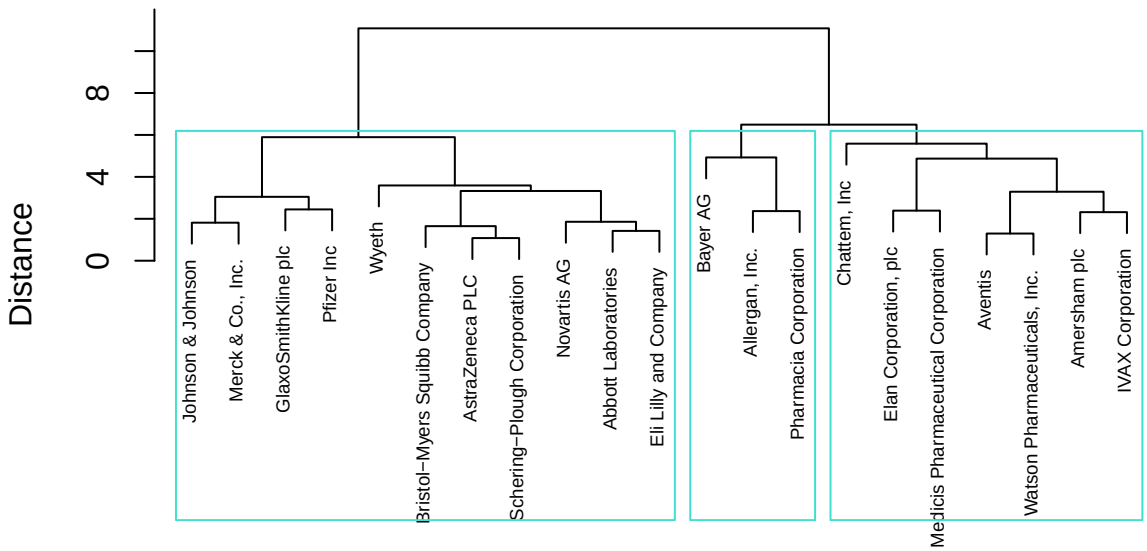
All nine variables were standardized, giving them equal weighting for the squared Euclidean distance. This is important as market cap is measured in Billions, where as other variables such as Beta and Asset Turnover are significantly smaller, which would have skewed the influence from market cap on the Euclidean Distance.

```r
# Calculate Euclidean distances
pharma_dist <- dist(pharma_scaled, method = "euclidean")
```

```r
# Perform hierarchical clustering using Ward's method
pharma_hc <- hclust(pharma_dist, method = "ward.D2")

# Create dendrogram
plot(pharma_hc,
     labels = pharma$Name,
     main = "Hierarchical Clustering of Pharmaceutical Firms",
     xlab = "Pharmaceutical Companies",
     ylab = "Distance",
     cex = 0.6)

# Highlight 3 clusters
rect.hclust(pharma_hc, k = 3, border = "turquoise")
```




```r
library(cluster)

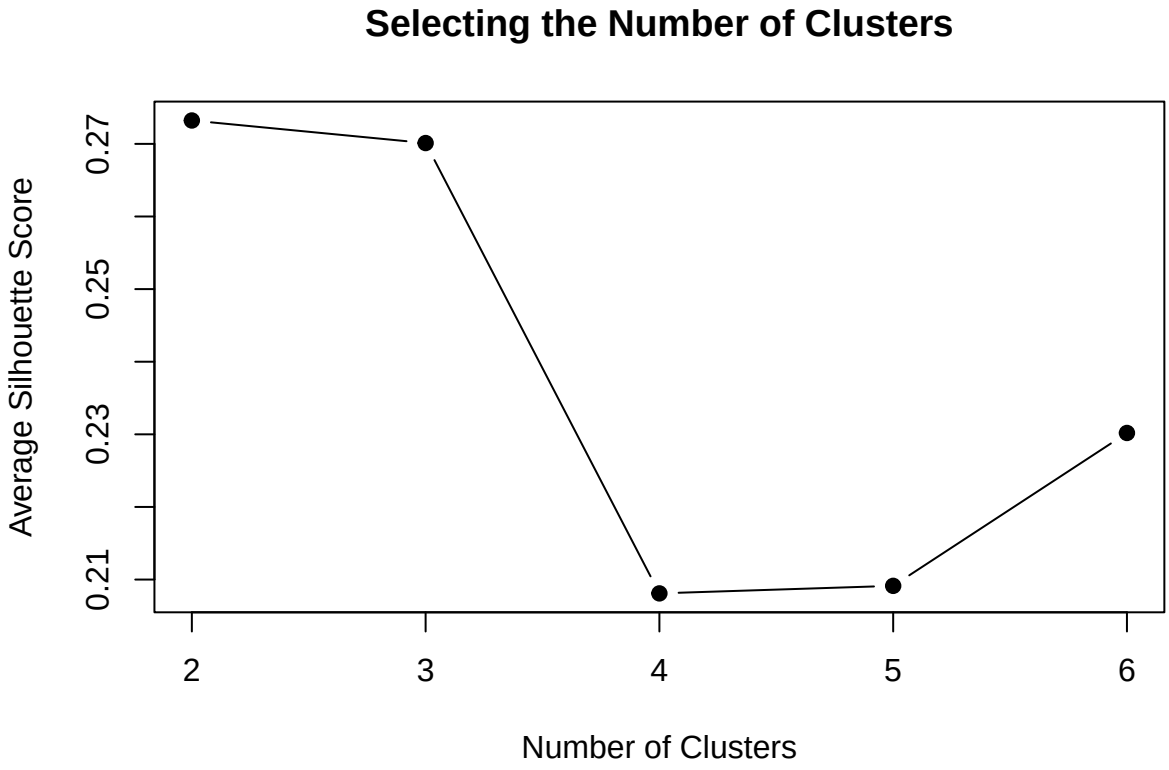
# Calculate silhouette scores for 2 through 6 clusters
sil_scores <- sapply(2:6, function(k) {
  clusters <- cutree(pharma_hc, k = k)
  sil <- silhouette(clusters, pharma_dist)
  mean(sil[, 3])
})

# Compare scores
data.frame(
  Number_of_Clusters = 2:6,
  Average_Silhouette = round(sil_scores, 3)
)
```

```
##   Number_of_Clusters Average_Silhouette
## 1                  2              0.273
## 2                  3              0.270
## 3                  4              0.208
## 4                  5              0.209
## 5                  6              0.230
```

```r
# Plot silhouette scores
plot(2:6, sil_scores,
     type = "b",
```

```
     pch = 19,
     xlab = "Number of Clusters",
     ylab = "Average Silhouette Score",
     main = "Selecting the Number of Clusters")
```



Part A ### For this Analysis, nine numerical financial variables were used and standardized with Z-scores to give equal weighting. The Euclidean Distance was then calculated and used to perform hierarchical clustering using Wards Method. This grouped the companies by minimizing increased within-cluster variation. A dendrogram was created to determine the appropriate number oc clusters, with the best coming to be 3 clusters, despite being 0.003 below 2, to give more meaningful insights into the financial distinctions in these companies. These companies were grouped into the three clusters, which contained 11, 3, and 7 each respectively. Although the silhouette score being .27 indicates a weak relationship, three clusters given the best idea to explain the differences in financial structure of the companies.

```
# Assign each company to one of the 3 clusters
pharma_clusters <- cutree(pharma_hc, k = 3)

# Add cluster assignments to the original dataset
pharma$Cluster <- pharma_clusters

# Display number of companies in each cluster
table(pharma$Cluster)
```

```
##
##  1  2  3
## 11  3  7
```

```r
# Calculate average financial values for each cluster
cluster_summary <- aggregate(
  pharma_numeric,
  by = list(Cluster = pharma_clusters),
  FUN = mean
)

# View results rounded to 2 decimal places
print(round(cluster_summary, 2))
```

```
##   Cluster Market_Cap Beta PE_Ratio   ROE   ROA Asset_Turnover Leverage
## 1       1      97.11 0.43    20.95 35.70 14.95           0.80     0.33
## 2       2      26.91 0.64    55.63 10.10  4.20           0.70     0.32
## 3       3       8.82 0.62    19.61 16.96  6.24           0.54     1.11
##   Rev_Growth Net_Profit_Margin
## 1      10.16             20.17
## 2       7.00              5.13
## 3      21.14             13.19
```

Part B

There are 3 Clusters of Companies

Cluster contains 11 companies, with the highest average market cap of $97.11 billion, return on equity of 35.7%, return on assets of 14.95%, and net profit margin of 20.17%. These 11 companies have the highest asset turnover at 0.8 and lowest beta of 0.43. This suggests these companies are larger, more established with bigger profitability, and lower sensitivity to market movement.

Cluster 2 is made of 3 companies that see the highest P/E ratio at 55.63, but the lowest return on equity, return on assets, and net profit margin at 10%, 4%, and 5% respectively. This suggests these companies have higher stock valuations, but weaker profitability. These 3 companies also have the lowest estimated revenue growth at 7%.

Cluster 3 is made of 7 companies, with the lowest average market capitalization at $8.82 billion, but the highest estimated revenue growth of 21.14%, and financial leverage of 1.11. They see a middle ground of net profit margin and return on equity, with the lowest asset turnover at 0.54. These companies are likely smaller but could see strong growth.

Overall, these clusters demonstrate differences in companies sizes, profitability, revenue, and financial leverage within the industry.

```r
# Analyst recommendations by cluster
table(pharma$Cluster, pharma$Median_Recommendation)
```

```
##
##     Hold Moderate Buy Moderate Sell Strong Buy
##   1    6            3             2          0
##   2    2            1             0          0
##   3    1            3             2          1
```

```r
# Compare Headquarter Locations
table(pharma$Cluster, pharma$Location)
```

```
##
##     CANADA FRANCE GERMANY IRELAND SWITZERLAND UK US
##   1      0      0       0       0           1  2  8
##   2      1      0       1       0           0  0  1
##   3      0      1       0       1           0  1  4
```

```r
# Compare Stock Exchanges Used
table(pharma$Cluster, pharma$Exchange)
```

```
##
##     AMEX NASDAQ NYSE
##   1    0      0   11
##   2    0      0    3
##   3    1      1    5
```

```r
# Create recommendation frequency table
rec_table <- table(
  pharma$Cluster,
  pharma$Median_Recommendation
)

# Calculate percentages within each cluster
round(prop.table(rec_table, margin = 1) * 100, 1)
```

```
##
##     Hold Moderate Buy Moderate Sell Strong Buy
##   1 54.5         27.3          18.2        0.0
##   2 66.7         33.3           0.0        0.0
##   3 14.3         42.9          28.6       14.3
```

Part C

Frequency Tables were used to compare the analyst recommendations, headquarter locations, and stock exchanges across the clusters.

Cluster 1 had the highest proportion of US-Based Companies with 8 of its 11 being headquartered in the States. Cluster 2 had a German and Canadian Company along with an American in its 3 companies. While Cluster 3 had a steady mix, with 3 International and 4 State headquartered companies shown. However, these locations did not show a pattern in seperating the Clusters.

This cluster analysis showcased some patterns among the variables that were not used to form the clusters. Clusters 1 and 2 are largely held by hold recommendations, representing 54.5% and 66.7% of their companies. Cluster 3 had a high proportion of Buy recommendations at 57.1%, combined with Moderate and Strong. Cluster 3 also had the highest Sell recommendation at 28.6%.

Every company from Cluster 1 and 2 was listed on the NYSE, while Cluster 3 contained 1 company from NASDAQ and AMEX respectively, and 5 companies on the NYSE. Overall, the recommendations and listings from stock exchanges showed differences between the clusters in some ways, and headquarter locations did not give away much relevant information.

Part D

Names for Each Cluster

Cluster 1 - Blue Chips

Cluster 2 - Speculative Firms

Cluster 3 - High-Growth Firms

These names were chosen due to their current and expected performance.

Cluster 1 has the highest average market cap, strongest profitability, as well as asset turnover. This means these are all likely blue chip corporations that are well known and established.

Cluster 2 companies have the highest P/E ratios, but lowest profitability. They likely have high valuations but have not performed yet, making them Speculative Firms.

Cluster 3 companies have low market cap, but see high estimated growth and leverage. These companies are expected to grow, and thus are called High-Growth Firms.

Overall Conclusions : The Silhouette showcased that 3 Clusters is the Optimal Value for this evaluation. The 3 Clusters were built based on characteristics from given pharmaceutical companies. Although these clusters were not strongly separated given the silhouette score of 0.27, it still gives useful insights into how these companies differ in size, profitability, and growth expectations.